Scatter Plots and Bivariate Association — Making and Qualifying Predictions

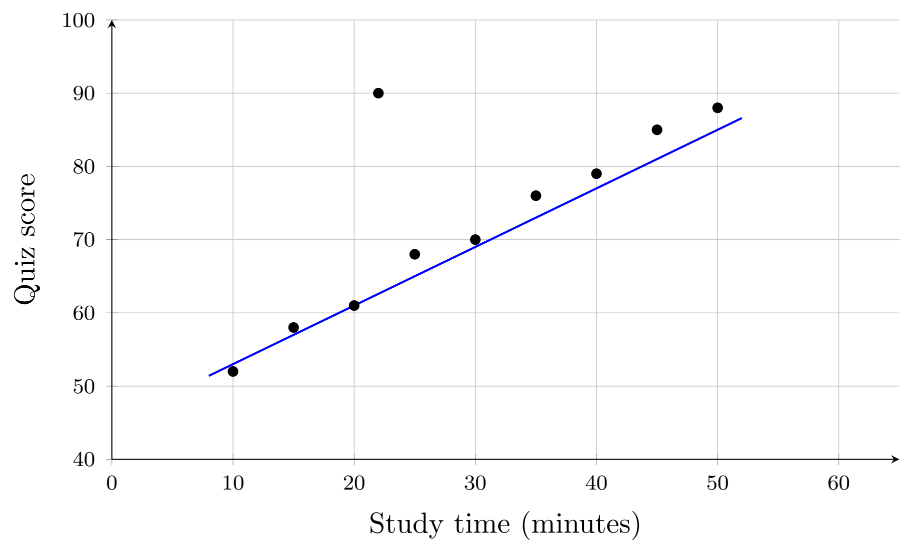
Reading association off a scatter plot, predicting from a trend line, and saying how far a prediction can honestly be trusted

Directions:

- **Scatter Plot A** below is shared by several problems. Read values from it where a problem says to; take everything else from the problem you are on.
- A prediction is only half the work. Every problem that asks you to *qualify* a prediction wants a reason in words: is the input inside the range the data covers, does one unusual point drive the result, and does association really mean cause?
- Round any decimal answer to the nearest hundredth unless a problem says otherwise.
- Show your work in the space under each problem, then write the final answer on the answer line.

Scatter Plot A — study time and quiz score for one class

each dot is one student; the straight line is the trend line $y = 0.8x + 45$



1. Use **Scatter Plot A**. One student studied for 45 minutes. What quiz score did that student earn?

Answer: ________________

2. Use **Scatter Plot A**. Which study time goes with the highest quiz score on the plot? Then say in one sentence whether that point fits the overall trend of the plot.

Answer: ________________

3. The trend line drawn on **Scatter Plot A** is $y = 0.8x + 45$, where x is study time in minutes and y is the predicted quiz score.

Use it to predict the score of a student who studies 30 minutes.

Answer: ________________

4. Use the same trend line.

(a) Predict the score of a student who studies 25 minutes.

(b) That student actually scored 68. Find the residual: the actual score minus the predicted score. Then say whether the line under-predicted or over-predicted this student.

Answer: ________________

5. A classmate looks at **Scatter Plot A** and says: *"This proves that studying longer causes higher quiz scores."*

Give one reason the plot cannot support the word *causes*, and name one other factor that could produce the same pattern.

Answer: ________________

6. Use **Scatter Plot A**. Find the mean quiz score of the ten students shown.

Then say whether the one unusual point pulls that mean up or down, and by roughly how much.

Answer: ____________

7. Use the trend line $y = 0.8x + 45$ again. Predict the quiz score of a student who studies for 60 minutes.

Answer: ____________

8. The study times on **Scatter Plot A** run from ten minutes to fifty minutes.

Using the words *interpolation* and *extrapolation*, explain why the sixty-minute prediction deserves less confidence than the thirty-minute one.

Answer: ________________

9. Use **Scatter Plot A**. How many points higher was the quiz score of the student who studied 50 minutes than the quiz score of the student who studied 10 minutes?

Answer: ________________

10. A second class fits its own trend line to its own scatter plot and gets $y = 1.3x + 27$. The first class's trend line is $y = 0.8x + 45$.

At what study time do the two trend lines predict exactly the same quiz score?

Answer: ________________

11. Remove the unusual point — the student who studied twenty-two minutes and scored ninety — and find the mean of the remaining nine quiz scores.

Compare it with the mean you found earlier, and say what that comparison shows about how much one unusual point can move a summary.

Answer: ________________

12. Challenge. The school newspaper wants to print this sentence: *"Students who study 75 minutes score about 105."*

Write a short paragraph judging the claim. Use the trend line, the range of study times the data actually covers, the unusual point, the highest score possible on the quiz, and the difference between association and causation. Say clearly what the data does support and what it does not.

Answer: ________________